TriAlign

Progress Overview & Blockers

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The Two Main Objectives

We compare extracted eligibility criteria in two steps:

1. Pair Identification

Which criteria could be compared?

- Extract atomic criteria from Trial A and Trial B.
- Search for candidate pairs by embedding (TF–IDF, MPNet or BioLORD)
- Human review to polish pairs

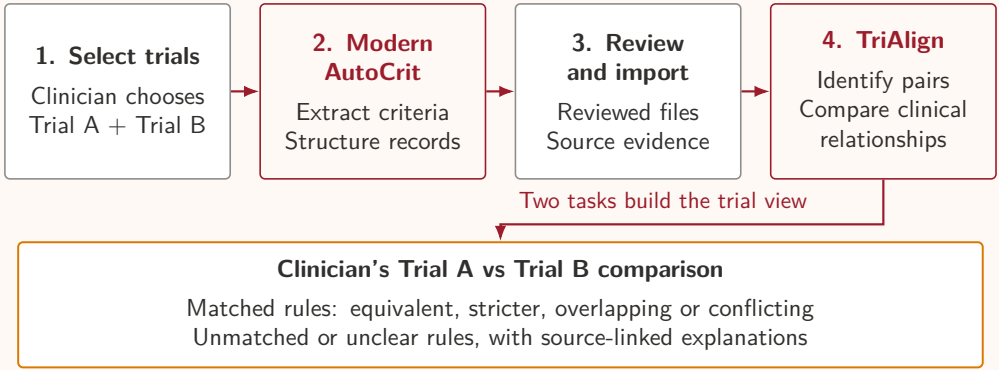
2. Relationship Comparison

How do those criteria relate clinically?

- Judge equivalence, restrictiveness, overlap and other relationships.
- Exact tools and LLM judge/critic, with an implemented learned relationship head.
- Output: relationship, evidence or review.

Example (rule summaries): A allows treated brain metastases without neurologic symptoms. B excludes all known brain metastases. **B is more restrictive**, despite high embedding similarity.

End-to-end Workflow



Current: Use criteria from Modern AutoCrit.

Planned: Atomic extraction directly within TriAlign, while retaining file import.

Current Dataset: Frozen and Reviewed

Data Source	Oncology and Cardiovascular Criteria via NCT API
Extraction	Modern AutoCrit produces atomic criteria with structured fields, 3100 criteria total
Validation	Reviewed and finalized in `trial100_v1.xlsx`
Frozen corpus	3,537 criteria across 100 trials, stored as Parquet
Evidence retained	original text, parent statements, trial IDs and source context

What the demo loads: select two trials from this frozen corpus, or import other extracted criterion records.

The demo compares existing extracted criteria. Direct extraction within TriAlign is planned. Criteria-from-protocol not added yet.

Dataset Visualization

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	source	parent_statement	criterion_type	domain	entity	comparator	value	unit	value_kind	lower_bound	lower_inclusive	upper_bound	upper_inclusive	negated	temporal	repetition	qualifier	logical_operator	review_status
2	Aged 18 - 80 years old	Aged 18 - 80 years old	inclusion	person	Age		18 to 80	years	interval	18	☑	80	☑	☐				standalone	corrected
3	Able to give informed c	Able to give informed consent	inclusion	person	Informed Consent		present		categorical					☐				standalone	corrected
4	Available for the whole	Available for the whole durat	inclusion	observation	availability for study duration		present		categorical					☐				standalone	corrected
5	Life expectancy \>2 yea	Life expectancy \>2 years, in t	inclusion	observation	Life Expectancy	>	2	years	numerical	2	☐			☐				standalone	corrected
6	Current systemic thera	Current systemic therapy for t	exclusion	drug	current systemic therapy for bladc		present		categorical					☐				standalone	corrected
7	History of malignancy o	History of malignancy of othe	exclusion	condition	malignancy of other organ system		present		categorical					☐	History of; within past 5 years; at least 24 months			or	corrected
8	History of malignancy o	History of malignancy of othe	inclusion	measurement	time since nephroureterecto	>=	24	months	numerical	24	☑			☐	after nephroureterectomy		exception to other-organ r	or	corrected
9	Eastern Cooperative On	Eastern Cooperative Oncolog	inclusion	observation	ECOG Performance Status		0 to 3		interval	0	☑	3	☑	☐				standalone	corrected
10	Serum creatinine ≤ 2.0 r	Serum creatinine ≤ 2.0 mg/dl.	inclusion	measurement	Serum Creatinine	<=	2	mg/dl	numerical			2	☑	☐				standalone	corrected
11	Subjects with detectabl	Subjects with detectable level	exclusion	condition	detectable levels of endogenous ar		present		categorical					☐			at the time of screening	standalone	corrected
12	In the absence of rapidl	In the absence of rapidly prog	inclusion	measurement	time since prior cytotoxic tr	>=	2	weeks	numerical	2	☑			☐	before study drug administration		in the absence of rapidly p	or	corrected
13	Histologically confirme	Histologically confirmed new	inclusion	condition	GBM		present		categorical					☐				standalone	corrected
14	Supratentorial tumor lc	Supratentorial tumor locatio	inclusion	condition	supratentorial tumor location		present		categorical					☐				standalone	corrected
15	Planned treatment with	Planned treatment with adju	inclusion	drug	TMZ		150 to 200	mg/m^2 dail	interval	150	☑	200	☑	☐				standalone	corrected
16	All patients with sufficie	All patients with sufficient tis	inclusion	observation	mgmt promoter methylation dete		present		categorical					☐	prior to enrollment			standalone	corrected
17	Karnofsky Performance	Karnofsky Performance Status	inclusion	measurement	Karnofsky Performance Stat	>=	70		numerical	70	☑			☐				standalone	corrected
18	Are surgically sterile, OI	Are surgically sterile, OR	inclusion	person	surgically sterile		present		categorical					☐				or	corrected
19	Unstable angina at regis	Unstable angina at registratio	exclusion	condition	unstable angina		present		categorical					☐	at registration			standalone	corrected
20	Transmural myocardial	Transmural myocardial infarc	exclusion	condition	transmural myocardial infarction		present		categorical					☐	within the last six months prior to registration			standalone	corrected
21	Patents treated on any (	Patents treated on any other t	exclusion	person	patients treated on any other ther		present		categorical					☐	within 30 days prior to registration			standalone	corrected
22	Inability to undergo MF	Inability to undergo MRI.	exclusion	device	inability to undergo mri		present		categorical					☐				standalone	corrected
23	Are postmenopausal fo	Are postmenopausal for at lea	inclusion	measurement	duration of postmenopausal	>=	1	year	numerical	1	☑			☐	before the screening visit		deterministically parsed v	or	corrected
24	Previously treated with	Previously treated with brent	exclusion	drug	brentuximab vedotin		present		categorical					☐				or	corrected
25	Previously treated with	Previously treated with brent	exclusion	drug	immune-oncology agents		present		categorical					☐				or	corrected
26	Previously treated with	Previously treated with brent	exclusion	procedure	allogeneic stem cell transplant		present		categorical					☐				or	corrected
27	Previously treated with	Previously treated with brent	exclusion	procedure	autologous stem cell transplant		present		categorical					☐				or	corrected
28	Documented history of a	Documented history of a care	exclusion	observation	cerebral vascular event		present		categorical					☐	history of			standalone	corrected
29	History of progressive n	History of progressive multifo	exclusion	condition	progressive multifocal leukoence		present		categorical					☐	History of			standalone	corrected
30	Hemoglobin ≥ 8.0 g/dl	Hemoglobin ≥ 8.0 g/dl	inclusion	measurement	Hemoglobin	>=	8	g/dl	numerical	8	☑			☐				standalone	corrected
31	Absolute neutrophil co	Absolute neutrophil count ≥ 1	inclusion	measurement	Absolute Neutrophil Count	>=	1000	mcL	numerical	1000	☑			☐				standalone	corrected
32	Platelet count ≥ 75,000	Platelet count ≥ 75,000/mcL	inclusion	measurement	Platelet Count	>=	75000	mcL	numerical	75000	☑			☐				standalone	corrected
33	Aspartate transaminase	Aspartate transaminase (AST)	inclusion	measurement	Aspartate Aminotransferase	<=	2.5		numerical			2.5	☑	☐				standalone	corrected
34	Alanine transaminase (A	Alanine transaminase (ALT) (SC	inclusion	measurement	Alanine Aminotransferase (A	<=	2.5		numerical			2.5	☑	☐				standalone	corrected
35	International normalize	International normalized ratio	inclusion	measurement	INR	<=	1.5		numerical			1.5	☑	☐				standalone	corrected
36	HBsAg negative, HBcAb,	HBsAg negative, HBcAb negat	inclusion	condition	hbsab positive		present		categorical					☐			applies to patients with HI	standalone	corrected

Each row retains the criterion text and parent statement alongside its role, clinical concept, numeric bounds, timing, qualifiers and review status.

Task 1: pair identification baselines

Jiatong: baseline pair ranking

- TF–IDF, MPNet and BioLORD score and rank candidate pairs.
- Cross-trial ranking with a shared neighbor format.
- Use the same frozen corpus of 3,537 criteria across 100 trials.

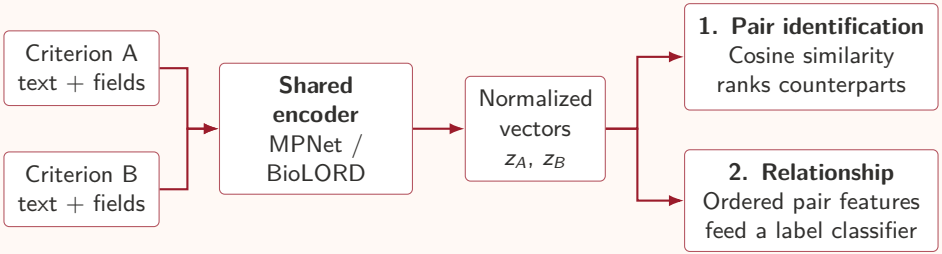
Current local demo

- Select TF–IDF, MPNet or BioLORD in the local interface.
- A clinical-concept gate and one-to-one assignment select candidate pairs.
- Selected candidate pairs pass to Task 2 for clinical relationship comparison.

TriAlign model: a shared criterion encoder supports pair identification and feeds an ordered relationship head.

Evaluate the two tasks separately: finding the right counterparts and assigning the right relationships.

TriAlign encoder architecture



Relationship input: $[z_A,\ z_B,\ z_A - z_B,\ |z_A - z_B|,\ z_A \odot z_B]$. Keeping A/B order supports directional labels.

Head: layer normalization, 256 hidden units by default, GELU, dropout and scores over eight configured labels.

Joint learning: counterpart similarity and relationship classification.

Future work: train and evaluate on human-validated pair labels.

Task 2: observed relationship comparison

16/16 pairs examined: 9 supported proposals; 7 need review

NCT01841736 · 104 criteria

"Women must not be pregnant or nursing"

NCT04310007 · 136 criteria

"Patients must not be pregnant or breast-feeding"

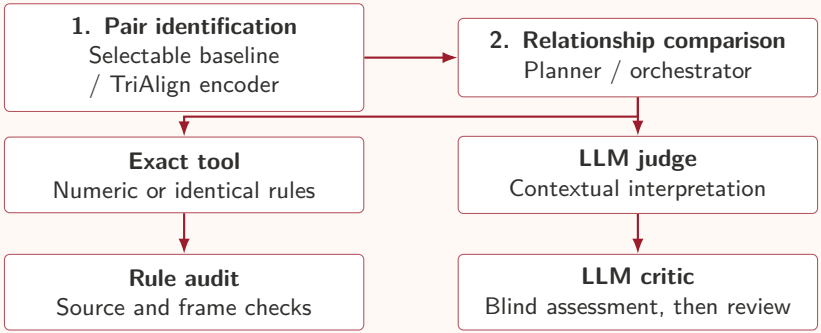
Pregnancy restrictions: proposed equivalent

Both trials exclude pregnant participants. Equivalence assumes comparable assessment at enrollment. Breastfeeding is compared separately.

Unresolved example: platelets. The critic rejected a directional label because Trial A's measurement timing was unspecified. Omitted timing does not establish a less restrictive rule.

One demonstration run. Agent acceptance is not clinical validation; accepted labels still require review.

The two tasks in the agentic workflow



Outcome: an evidence-supported proposal or an unresolved case for human review.

The orchestrator can request a learned relationship proposal before judge/critic review. It allows one revision, then finalizes or escalates.

Skills, tools and hooks give the agents boundaries

Component	Role in the working system
Skills	Versioned instructions for numeric comparison, clinical context and source problems. The planner selects a skill.
Tools	Inspect the comparison frame, compare supported constraints and collect source citations. Only permitted tools run.
Hooks	Validate inputs and exact source quotes, check limits and cancellation, and gate the final result.
Orchestrator	Coordinates tool calls, judge, critic and revision within pair, call and time limits.

Clinician-facing output: source excerpts, rationale, critic feedback, executed trace and saved reviewer decisions.

The critic first assesses sources blind, then reviews the judge. Models are separately configurable; agreement is not clinical validation.

Closest related methods and benchmarks

Related work	Overlap with TriAlign	Planned comparison
CReSE	Criterion embeddings and clinical relevance	Task 1: rank counterpart criteria.
EligMeta	Eligibility alignment to a target trial	Task 2: compare supported criterion differences.
NLI4CT (benchmark)	Clinical-trial entailment and contradiction	Task 2: stress-test reasoning and consistency.
AMEND++ (benchmark)	Eligibility amendment prediction and version histories	Adapt substantive edits for semantic-change tests.

Planned comparisons, subject to artifact access. Native outputs differ from TriAlign's relationship labels, so matched cases or explicit adaptations are required.

Current challenges and limitations

- Observed failures include source-quotation mismatches and unsupported timing assumptions, including in some accepted overlap proposals.
- Judge and critic can share mistakes. Human review remains necessary.
- No retrieved candidate does not prove that a requirement is absent.
- Family-aware partitions are frozen. Human-validated pair labels and held-out clinical evaluation remain pending.
- The encoder and relationship head are implemented. Validated training, baseline comparisons and uncertainty calibration are future work.

The UI runs locally. Contextual model calls still need network access and have API costs and latency.

Next steps before the final presentation

Task	Deliverable
Human ground truth	Finish pair review, resolve disagreements and obtain independent second reviews.
Held-out pair data	Construct and label validation/test pairs within the frozen trial-family partitions.
Model training	Fine-tune the shared encoder and relationship head using human-validated training labels.
Clinical evaluation	Compare pair identification and relationship prediction separately against baselines. Report class errors and abstention.
Local demonstration	Refine import/review flows, measure latency and rehearse locally.

Pair construction and family-aware sampling

Initial pilot: 75 cross-trial pairs

Sampling stratum	Pairs
Same entity, structured differences	25
Other same-entity pairs	25
Lexical neighbors	15
Random controls	10

Frozen family-aware partitions

- Audited trial identity, lineage and copied criteria across 100 trials.
- Reserved pilot trials for development; 21 trials excluded from held-out testing.
- Keep linked trials together: 68 training, 15 validation, 15 test and 2 quarantined trials.

Current review: 375 pairs in the workbook, including the initial pilot. Human validation remains in progress.

Next sampling: build separate validation/test review packets using same-concept and baseline-ranked candidates, difficult scope/negation cases and random controls.